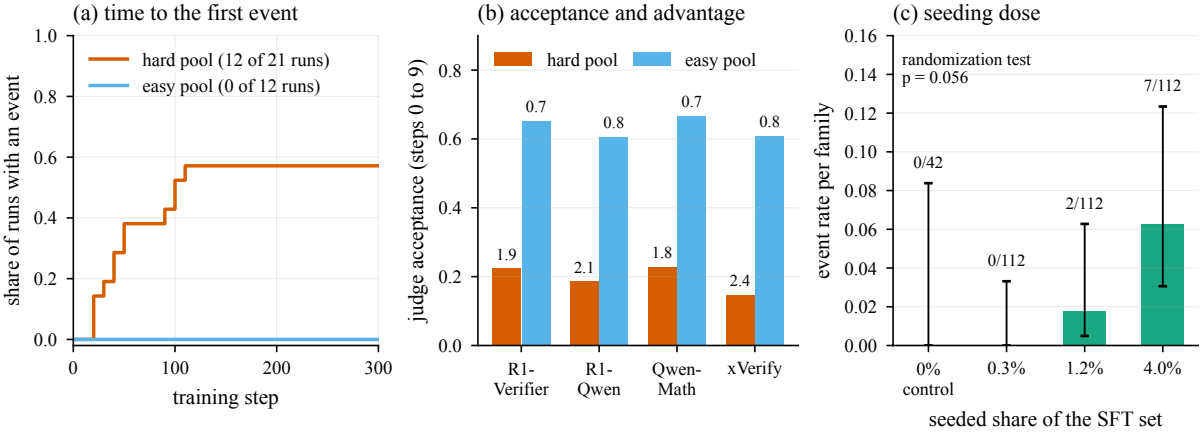
(a) time to the first event
hard pool (12 of 21 runs)
easy pool (0 of 12 runs)
share of runs with an event
training step
(b) acceptance and advantage
hard pool
easy pool
judge acceptance (steps 0 to 9)
0.7
0.8
0.7
0.8
1.9
2.1
1.8
2.4
R1-Verifier
R1-Qwen
Qwen-Math
xVerify
(c) seeding dose
randomization test
p = 0.056
event rate per family
0/42
0/112
2/112
7/112
0% control
0.3%
1.2%
4.0%
seeded share of the SFT set